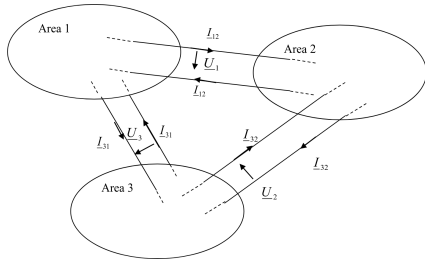


Exercises

Exercise 1: Power-flow sign convention

Data:

$$\begin{aligned} \underline{U}_1 &= 76\angle 4^\circ \text{ kV}, & \underline{U}_2 &= 76\angle 8^\circ \text{ kV}, & \underline{U}_3 &= 76\angle -11^\circ \text{ kV}, \\ \underline{I}_{12} &= 450\angle -15^\circ \text{ A}, & \underline{I}_{31} &= 300\angle 18^\circ \text{ A}, & \underline{I}_{32} &= 500\angle -11^\circ \text{ A} \end{aligned}$$



- Which areas provide active power? How much?
- Which areas receive reactive power? How much?

$$\underline{S} = \underline{U} \cdot \underline{I}^* = \underline{U} + \underline{I}\angle(-\theta^\circ)$$

$$\begin{aligned} \underline{S}_1 &= \underline{U}_1 \underline{I}_{31}^* = 76\angle 4^\circ \cdot 450\angle 15^\circ - 76\angle -11^\circ \cdot 300\angle -18^\circ \\ &= 12.4 + j22.2 \text{ [MW]} = 12.4 \text{ [MW]} + 22.2 \text{ [MVar]} \\ \underline{S}_2 &= \underline{U}_2 \underline{I}_{32}^* - \underline{U}_1 \underline{I}_{12}^* = 3.59 \text{ [MW]} + j1.24 \text{ [MVar]} \\ \underline{S}_3 &= \underline{U}_3 \underline{I}_{31}^* - \underline{U}_2 \underline{I}_{32}^* = -15.99 \text{ [MW]} - j23.43 \text{ [MVar]} \end{aligned}$$

Conventionally, $S, Q > 0$: provider, $S, Q < 0$: receiver

- $\underline{S}_1$: 2: provider, 12.4 MW and 3.59 MW
- $\underline{S}_3$: receiver, 23.43 MVar

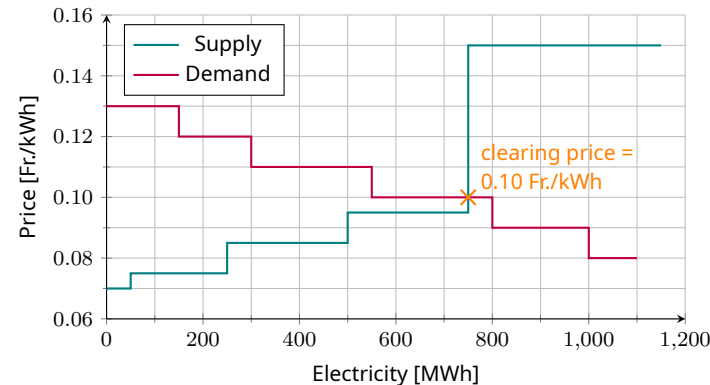
Exercise 2: Merit-order pricing

Sorted list of B1–B6 consumer offers:

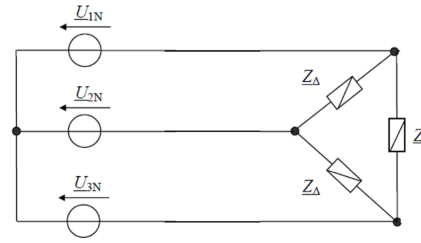
B1	B2	B3	B4	B5	B6
100 MWh 0.08 Fr./kWh	200 MWh 0.09 Fr./kWh	250 MWh 0.10 Fr./kWh	250 MWh 0.11 Fr./kWh	150 MWh 0.12 Fr./kWh	150 MWh 0.13 Fr./kWh

Sorted list of offers of electricity sellers S1–S5:

S1	S2	S3	S4	S5
50 MWh 0.07 Fr./kWh	200 MWh 0.075 Fr./kWh	250 MWh 0.085 Fr./kWh	250 MWh 0.095 Fr./kWh	400 MWh 0.15 Fr./kWh



Exercise 3: Three-phase power



Data:

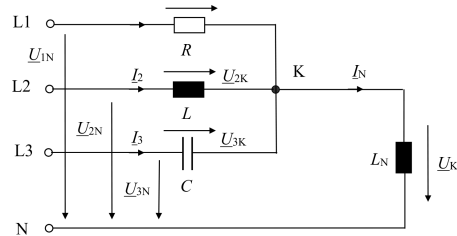
$$\begin{aligned} \underline{U}_{1N} &= 110\text{ V}\angle 0^\circ, & \underline{U}_{2N} &= 230\text{ V}\angle -120^\circ, & \underline{U}_{3N} &= 230\text{ V}\angle 120^\circ, \\ \underline{Z}_\Delta &= (30 + j25) \Omega \end{aligned}$$

Calculate the complex power $\underline{S}$ and the power factor λ

$$\underline{S} = \frac{\sum_i |\underline{U}_i|^2}{\underline{Z}_\Delta^*}$$

$$\begin{aligned} \underline{U}_{12} &= \underline{U}_{1N} - \underline{U}_{2N} = 110\angle 0^\circ \text{ V} - 230\angle -120^\circ \text{ V} = 300.5\angle 41.52^\circ \text{ V} \\ \underline{U}_{23} &= \underline{U}_{2N} - \underline{U}_{3N} = 230\angle -120^\circ \text{ V} - 230\angle 120^\circ \text{ V} = 398.5\angle -90^\circ \text{ V} \\ \underline{U}_{31} &= \underline{U}_{3N} - \underline{U}_{1N} = 230\angle 120^\circ \text{ V} - 110\angle 0^\circ \text{ V} = 300.5\angle 138.48^\circ \text{ V} \\ \underline{S} &= \frac{|\underline{U}_{12}|^2 + |\underline{U}_{23}|^2 + |\underline{U}_{31}|^2}{\underline{Z}_\Delta^*} = \frac{(300.5\text{ V})^2 + (300.5\text{ V})^2 + (398.5\text{ V})^2}{39.05\angle -39.81^\circ \Omega} \\ &= 6.68 + j5.56 \text{ kVA} = 8.69\angle 39.81^\circ \text{ kW} \\ \lambda &= \cos(39.81^\circ) = 0.768 \end{aligned}$$

Exercise 4: 4-wire system



Data:

$$\begin{aligned} \underline{U}_{1N} &= 220\text{ V}\angle 0^\circ, & R &= 100 \Omega, \\ \underline{U}_{2N} &= 220\text{ V}\angle -120^\circ, & L &= 160 \text{ mH}, \\ \underline{U}_{3N} &= 220\text{ V}\angle 120^\circ, & C &= 40 \mu\text{F}, \\ f &= 50 \text{ Hz}, & L_N &= 500 \text{ mH}. \end{aligned}$$

Calculate: $\underline{I}_1, \underline{I}_2, \underline{I}_3, \underline{I}_N$, and $\underline{U}_{1K}, \underline{U}_{2K}, \underline{U}_{3K}, \underline{U}_{KN}$.

$$\begin{aligned} \underline{U}_{KN} &= \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N} \\ \underline{U}_{iK} &= \underline{U}_{iN} - \underline{U}_{KN} \quad ; \quad \underline{I}_i = \underline{U}_{iK} \cdot \underline{Y}_i \end{aligned}$$

Admittances:

$$\begin{aligned} \underline{Y}_1 &= \underline{Y}_R = \frac{1}{R} = 0.010\angle 0^\circ \text{ S}, \\ \underline{Y}_2 &= \underline{Y}_L = \frac{1}{\omega L} \angle -90^\circ = 0.0199\angle -90^\circ \text{ S}, \\ \underline{Y}_3 &= \underline{Y}_C = \omega C \angle 90^\circ = 0.0126\angle 90^\circ \text{ S}, \\ \underline{Y}_N &= \underline{Y}_L = \frac{1}{\omega L_N} \angle -90^\circ = 0.00637\angle -90^\circ \text{ S}. \end{aligned}$$

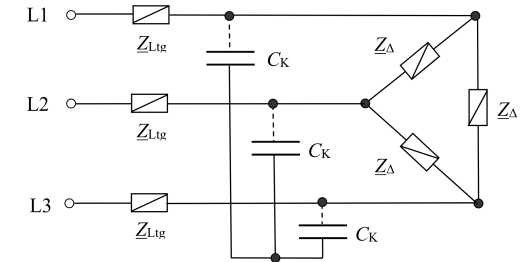
Voltages:

$$\begin{aligned} \underline{U}_{KN} &= \frac{\underline{Y}_1 \underline{U}_{1N} + \underline{Y}_2 \underline{U}_{2N} + \underline{Y}_3 \underline{U}_{3N}}{\underline{Y}_1 + \underline{Y}_2 + \underline{Y}_3 + \underline{Y}_N} = 238.4\angle -137.5^\circ \text{ V}, \\ \underline{U}_{1K} &= \underline{U}_{1N} - \underline{U}_{KN} = 427.2\angle 22^\circ \text{ V}, \\ \underline{U}_{2K} &= \underline{U}_{2N} - \underline{U}_{KN} = 71.9\angle -24^\circ \text{ V}, \\ \underline{U}_{3K} &= \underline{U}_{3N} - \underline{U}_{KN} = 357.8\angle 79.4^\circ \text{ V}. \end{aligned}$$

Currents:

$$\begin{aligned} \underline{I}_1 &= \underline{U}_{1K} \underline{Y}_1 = 4.27\angle 22^\circ \text{ A}, \\ \underline{I}_2 &= \underline{U}_{2K} \underline{Y}_2 = 1.43\angle -114^\circ \text{ A}, \\ \underline{I}_3 &= \underline{U}_{3K} \underline{Y}_3 = 4.51\angle 169.4^\circ \text{ A}, \\ \underline{I}_N &= \underline{U}_{KN} \underline{Y}_N = 1.52\angle 132.5^\circ \text{ A}. \end{aligned}$$

Exercise 5: Three-phase systems



Data:

$$U_{LL} = 400\text{ V}/50\text{ Hz} \quad ; \quad \underline{Z}_{Ltg} = (1 + j2) \Omega \quad ; \quad \underline{Z}_\Delta = 30\angle 60^\circ$$

- Calculate the line currents $\underline{I}$ and the voltages $\underline{U}_\Delta$ across the loads $\underline{Z}_\Delta$ without the capacitor bank connected.
- Calculate the capacitance C_K of the capacitor bank for reactive current compensation of the loads $\underline{Z}_\Delta$.
- Determine the line currents $\underline{I}$ in the compensated case.

a) Without compensation:

$$\begin{aligned} \underline{I}_L &= \frac{U_{ph}}{\underline{Z}_{Ltg} + \underline{Z}_Y} = \frac{U_{LL}}{\sqrt{3}} \cdot \frac{1}{\underline{Z}_{Ltg} + \frac{\underline{Z}_\Delta}{3}} \\ \underline{U}_\Delta &= \sqrt{3} \underline{U}_Y = \sqrt{3} (\underline{I}_L \underline{Z}_Y) = \sqrt{3} \left(\underline{I}_L \cdot \frac{\underline{Z}_\Delta}{3} \right) \end{aligned}$$

$$I_1 = \frac{400V}{\sqrt{3}} \cdot \frac{1}{(1+j2)\Omega + \frac{30\angle 60^\circ}{3}} = 18.88\angle -60.6^\circ \text{ A}$$

$$I_2 = I_1 - 120^\circ = 18.88\angle 179.4^\circ$$

$$I_3 = I_1 + 120^\circ = 18.88\angle 59.4^\circ$$

$$U_\Delta = \sqrt{3} \left(18.88\angle -60.6^\circ \cdot \frac{30\angle 60^\circ}{3} \right) = 327\angle -0.63^\circ \text{ [V]}$$

$$U_{12,\Delta} = U_\Delta + 30^\circ = 327\angle 29.7^\circ \text{ [V]}$$

$$U_{23,\Delta} = U_\Delta - 120^\circ = 327\angle -90.63^\circ \text{ [V]}$$

$$U_{31,\Delta} = U_\Delta + 120^\circ = 327\angle 149.37^\circ \text{ [V]}$$

$$Z_\Delta = \phi = 30\angle 60^\circ \Omega$$

b) Compensation capacitance:

$$Y_C = j\omega C_k \quad ; \quad Y_Y = \frac{1}{Z_Y} \quad ; \quad \text{Im}(Y_C + Y_Y) = 0$$

$$Y_Y = \frac{3}{Z_\Delta} = 0.05 - j0.0866 \text{ S} \rightarrow \text{Im}(Y_Y) = -0.0866 \text{ S}$$

$$\text{Im}(Y_C) = -Y_Y = 0.0866 \text{ S}$$

$$0.0866 = \omega C_k \Rightarrow C_k = \frac{0.0866}{2\pi \cdot 50 \text{ Hz}} = 275.8 \text{ } [\mu\text{F}]$$

c) With compensation:

$$Y_{\text{load}} = \text{Re}(Y_Y) \quad ; \quad Z_{\text{load}} = \frac{1}{Y_{\text{load}}} \quad ; \quad Z_{\text{tot}} = Z_{\text{load}} + Z_{\text{Ltg}}$$

$$Z_{\text{tot}} = 20\Omega + (1+j2)\Omega = 21 + j2\Omega$$

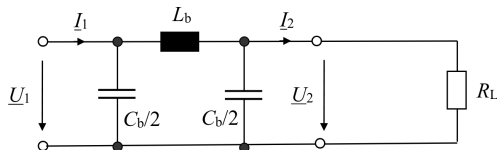
$$I_L = \frac{U_{LL}}{Z_{\text{tot}}} \rightarrow I_1 = \frac{400 \text{ [V]}}{\sqrt{3} \cdot 21 + j2\Omega} = 10.95\angle -5.44^\circ \text{ [A]}$$

$$I_2 = I_1 - 120^\circ = 10.95\angle -125.44 \text{ [A]}$$

$$I_3 = I_1 + 120^\circ = 10.95\angle 114.56 \text{ [A]}$$

Exercise 6: Surge impedance loading

Equivalent line model of a lossless line with load R_L :



Determine R_L so that the lossless line is optimal operation. **Data:**

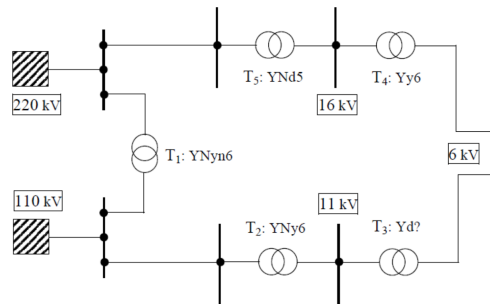
$$L'_b = 1.3 \text{ [mH/(km and phase)]} \quad ; \quad C'_b = 9.6 \text{ [nF/(km and phase)]}$$

$$l = 250 \text{ [km (length of the line)]}$$

$$Z_C = \sqrt{\frac{L'_b}{C'_b}} = \sqrt{\frac{L_b}{C_b}} \quad ; \quad L_b = L'_b \cdot l \quad ; \quad C_b = C'_b \cdot l$$

$$R_L = Z_C = \sqrt{\frac{1.3\text{E}-3}{9.6\text{E}-9}} = 368 \Omega$$

Exercise 7: Transformers 1



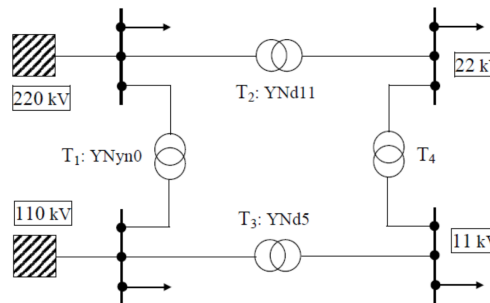
Determine the index number of transformer T_3 .

$$(T_5 + T_4) \bmod 12 = (T_3 + T_2 + T_1) \bmod 12$$

$$11 \bmod 12 = (T_3 + 12) \bmod 12$$

$$(11 - 12) \bmod 12 \equiv T_3 \rightarrow -1 \bmod 12 \equiv T_3 \rightarrow T_3 = 11$$

Exercise 8: Transformer 2



Determine a suitable transformer configuration for T_4 .

$$T_2 + T_4 = T_3 + T_2 \Rightarrow 11 + T_4 = 5$$

$$11 + T_4 \equiv 5 \bmod 12 \rightarrow T_4 = 6$$

Suitable transformers: Dd6, Yy6, Dz6.

Exercise 9: Active and reactive power of a synchronous generator

A three-phase synchronous generator (star connected) is built as a non-salient pole machine. It operates at constant voltage and frequency.

Data:

$$S_N = 400 \text{ MVA} \quad ; \quad U_N = 20 \text{ kV} / 50 \text{ Hz} \quad ; \quad X_d = 2 \text{ p.u.}$$

$$E = 25 \text{ kV} \quad ; \quad \delta = 30^\circ$$

U_N , unless otherwise specified, is line-to-line voltage.

Determine the active and reactive power of the generator. Is the machine capacitive or inductive?

$$P = \frac{U \cdot U_p}{X_d} \sin(\vartheta) \quad ; \quad P_{3p} = 3 \cdot P$$

$$Q = \frac{U \cdot U_p}{X_d} \cos(\vartheta) - \frac{U^2}{X_d} \quad ; \quad Q_{3p} = 3 \cdot Q$$

$$P = \frac{20\text{E3 [V]} \cdot 25\text{E3 [V]}}{\sqrt{3} \cdot 2 \text{ [p.u.]}} \sin(30^\circ) = 72.2 \text{ [MW]}$$

$$P_{3p} = 3 \cdot 72.2 = 216.6 \text{ [MW]}$$

$$Q = \frac{20\text{E3 [V]} \cdot 25\text{E3 [V]}}{\sqrt{3} \cdot 2 \text{ [p.u.]}} \cos(30^\circ) - \frac{\left(\frac{20\text{E3 [V]}}{\sqrt{3}} \right)^2}{2 \text{ p.u.}} = 58.33 \text{ [MVA]}$$

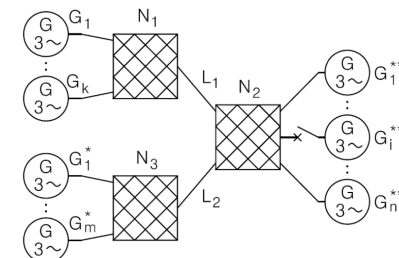
$$P_{3p} = 3 \cdot 58.33 = 175 \text{ [MVA]}$$

Exercise 10: Interconnected control areas

Three control areas N_1 - N_2 - N_3 are connected by L_1 and L_2 , initially without power exchange. A 100 MW plant fails in N_2 .

$$f_N = 50 \text{ Hz} \quad ; \quad K_{N1} = 400 \text{ MW/Hz} \quad ; \quad K_{N2} = K_{N3} = 500 \text{ MW/Hz}$$

- What frequency will occur in the control areas after the primary controllers have been activated?
- What power is then exchanged between the areas?
- What input signals ΔP_R are given to the secondary controllers of the three areas?
- When will the power exchange on L1 and L2 return to normal?



a) Frequency shift:

$$\Delta P_i = -\sum_i (K_{Ni}) \Delta f \Leftrightarrow \Delta f = \frac{\Delta P}{-(\sum_i K_{Ni})}$$

$$\Delta f = \frac{100 \text{ [MW]}}{-(400 + 500 + 500) \text{ [MW/Hz]}} = -0.0714 \text{ [Hz]}$$

b) Power exchanged:

$$\Delta P_i = -K_{Ni} \Delta f$$

$$\Delta P_1 = -400 \text{ [MW/Hz]} \cdot (-0.0714) \text{ [Hz]} = 28.56 \text{ [MW]}$$

$$\Delta P_2 = \Delta P_3 = -500 \text{ [MW/Hz]} \cdot (-0.0714) \text{ [Hz]} = 35.7 \text{ [MW]}$$

c) Secondary controllers:

$$\Delta P_{Ri} = \Delta P_i - \Delta P_{Ki}$$

$$\Delta P_{R1} = \Delta P_1 + K_{N1} \Delta f = 28.56 \text{ [MW]} - 28.56 \text{ [MW]} = 0 \text{ [MW]}$$

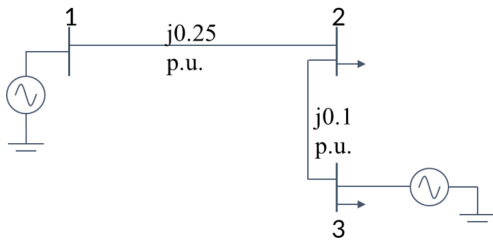
$$\Delta P_{R2} = \Delta P_3 + K_{N3} \Delta f = 35.7 \text{ [MW]} - 35.7 \text{ [MW]} = 0 \text{ [MW]}$$

$$\Delta P_{R3} = \Delta P_2 + K_{N2} \Delta f = -(\Delta P_1 + \Delta P_3) + K_{N2} \Delta f = -(28.56 + 37.5) \text{ [MW]} + 500 \cdot (-0.0714) \text{ [MW]} = -100 \text{ [MW]}$$

d) Time period: Between 5 to 15 minutes.

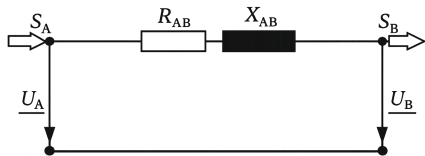
Exercise 11: Ybus

Form the Ybus of the following system:



$$\begin{cases} I_1 = \frac{1}{j0.25} (v_1 - v_2) \\ I_2 = \frac{1}{j0.1} (v_2 - v_3) + \frac{1}{j0.25} (v_2 - v_1) \\ I_3 = \frac{1}{j0.1} (v_3 - v_2) \end{cases} \quad \begin{cases} I_1 = -j4v_1 + j4v_2 \\ I_2 = -j14v_2 + j10v_3 + j4v_1 \\ I_3 = -j10v_3 + j10v_2 \end{cases}$$

$$\begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -j4 & j4 & 0 \\ j4 & -j14 & j10 \\ 0 & j10 & -j10 \end{bmatrix} \cdot \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

Exercise 12a: Two-node system (P_B and Q_B)Data:

$$\underline{U}_A = 400 \angle 0^\circ [\text{V}], \quad Z_{AB} = 0.1 + j0.1 [\Omega], \quad \underline{U}_B = 396.3 e^{j(-0.1774^\circ)} [\text{V}]$$

Calculate P_B and Q_B . Is the node inductive or capacitive?

$$\begin{aligned} Z_{AB} &= R_{AB} + jX_{AB} = |Z_{AB}| e^{j\theta} \\ P_B &= \frac{|U_A| |U_B|}{|Z_{AB}|} \cos(\theta + \delta) - \frac{|U_B|^2}{|Z_{AB}|} \cos(\theta) \\ Q_B &= \frac{|U_A| |U_B|}{|Z_{AB}|} \sin(\theta + \delta) - \frac{|U_B|^2}{|Z_{AB}|} \sin(\theta) \end{aligned}$$

$$Z_{AB} = 0.1414 e^{j45^\circ} [\Omega]$$

$$P_B = \frac{400 \cdot 396.3}{0.1414} \cos(45^\circ - 0.1774^\circ) - \frac{396.3^2}{0.1414} \cos(45^\circ) = 9815 [\text{W}]$$

$$Q_B = \frac{400 \cdot 396.3}{0.1414} \sin(45^\circ - 0.1774^\circ) - \frac{396.3^2}{0.1414} \sin(45^\circ) = 4908 [\text{Var}]$$

 $Q_B > 0 \rightarrow$ Node B consumes reactive power $\rightarrow$ Inductive system**Exercise 12b: Two-node system (Iterative process for U_B)**Data:

$$U_A = 400 [\text{V}]; \quad Z_{AB} = 0.1 + j0.1 [\Omega]; \quad P_B = 9815 [\text{W}]; \quad Q_B = 4908 [\text{Var}]$$

Calculate $\underline{U}_B$ (magnitude and phase angle δ).Step 1) Initialize $i = 0$ and $\underline{U}_B^{(0)} = \underline{U}_A$.

$$\text{Step 2) Calculate the current: } \underline{I}_B^{(i)} = \left\{ \frac{P_B + jQ_B}{\underline{U}_B^{(i)}} \right\}^*$$

$$\text{Step 3) Update the voltage: } \underline{U}_B^{(i+1)} = \underline{U}_A - Z_{AB} \underline{I}_B^{(i)}.$$

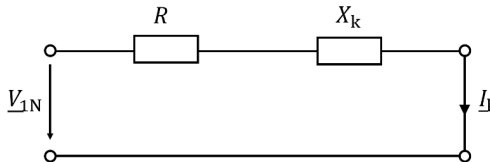
Step 4) Check convergence: $\left| \underline{U}_B^{(i+1)} \right| - \left| \underline{U}_B^{(i)} \right| < \varepsilon$; $\varepsilon = 1\text{E} - 5$.
Otherwise, set $i = i + 1$ and repeat from Step 2.

Calculator iteration:Set Ans = $\underline{U}_A$, then repeatedly enter

$$\underline{U}_A - (Z_{AB}) \frac{P_B - jQ_B}{\text{CONJ}(\text{Ans})}$$

until the result no longer changes. The final result is $\underline{U}_B$.Result:

$$\underline{U}_B = 396.28 \angle -0.18^\circ [\text{V}] = 396.28 - j1.23 [\text{V}]$$

Exercise 13: Short circuitsData:

$$V_L = 20 [\text{kV}] \quad ; \quad f = 50 [\text{Hz}] \quad ; \quad I_k = 300 [\text{A}] \quad ; \quad R = 12 [\Omega]$$

- a) Determine the value of the leakage reactance X_k
b) Calculate the short-circuit current at time $t = 0.03$ s.
Assume that the short occurred at $t = 0$ s.

$$\begin{aligned} i(t) &= i_s(t) + i_t(t) \\ i_s(t) &= \frac{\sqrt{2}V_S}{|Z|} \sin(\omega t + \alpha - \Theta) \\ i_t(t) &= -i_s(0) \cdot e^{-(R/L)t} = \frac{\sqrt{2}V_S}{|Z|} \sin(\Theta - \alpha) \cdot e^{-(R/L)t} \\ |Z| &= \sqrt{R^2 + \omega^2 L^2} \angle \left(\Theta = \tan^{-1} \left(\frac{\omega L}{R} \right) \right) \end{aligned}$$

a) Leakage reactance:

$$\begin{aligned} V_{1N} &= \frac{V_L}{\sqrt{3}} \quad ; \quad I_k = \frac{V_{1N}}{|Z|} \Rightarrow |Z| = \frac{V_L}{\sqrt{3}I_k} \\ |Z| &= \sqrt{R^2 + X_k^2} \Rightarrow X_k = \sqrt{|Z|^2 - R^2} = \sqrt{\left(\frac{V_L}{\sqrt{3}I_k} \right)^2 - R^2} \\ &= \sqrt{\left(\frac{20\text{E}3 [\text{V}]}{\sqrt{3} \cdot 300 [\text{A}]} \right)^2 - (12 [\Omega])^2} = 36.57 [\Omega] \end{aligned}$$

b) Short circuit:

$$X_k = 2\pi f K \Rightarrow K = \frac{X_k}{2\pi f} = \frac{36.57 [\Omega]}{2\pi \cdot 50 [\text{Hz}]} = 0.1164 [\text{H}]$$

$$i_t(0) = -i_s(0) e^{-(R/K)t}$$

$$i_t(0.03) = \sqrt{2}I_k \sin(\Theta) e^{-(12/0.1164) \cdot 0.03}$$

$$= \sqrt{2}I_k \sin \left(\arctan \left(\frac{X_k}{R} \right) \right) e^{-(12/0.1164) \cdot 0.03}$$

$$= \sqrt{2} \cdot 300 [\text{A}] \sin \left(\arctan \left(\frac{36.57 [\Omega]}{12 [\Omega]} \right) \right) e^{-(12/0.1164) \cdot 0.03}$$

$$= 18.29 [\text{A}]$$

$$\begin{aligned} i_s(0.03) &= \sqrt{2}I_k \sin \left(2\pi f \cdot 0.03 [\text{s}] - \arctan \left(\frac{X_k}{R} \right) \right) \\ &= \sqrt{2} \cdot 300 [\text{A}] \sin \left(2\pi \cdot 50 [\text{Hz}] \cdot 0.03 [\text{s}] - \arctan \left(\frac{0.1164 [\text{H}]}{12 [\Omega]} \right) \right) \\ &= 403.12 [\text{A}] \end{aligned}$$

$$i(t) = i_t(t) + i_s(t) = 403.12 [\text{A}] + 18.29 [\text{A}] = 421.41 [\text{A}]$$